

Learning TikZ

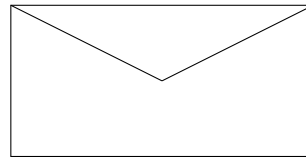
1 Inline Graphics

Inline small rectangle: $\square$, this is a small $\rule{1cm}{0.4pt}$ Line $\diagup$

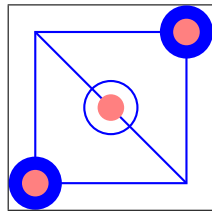
For multiple objects: $\square\square\square$

Some textual content $\odot$ Other textual content.

Now centering the circle $\bigcirc$ to the surrounding text.

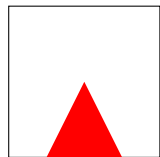


The following draw is a 4×2 Crossed rectangle:



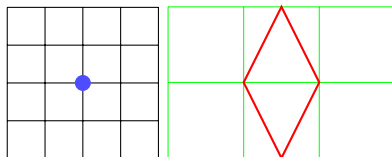
Some textual Content.

1.2 The Bounding Box



Gravity or gravitation, is a natural phenomenon by which all things with mass or energy-including planets, stars, galaxies, and even light are brought toward (or gravitate toward) one another. On Earth, gravity gives weight to physical objects, and the Moon's gravity causes the ocean tides.

2 Placing Grids

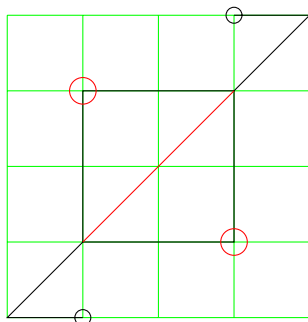


3 Constructing Paths

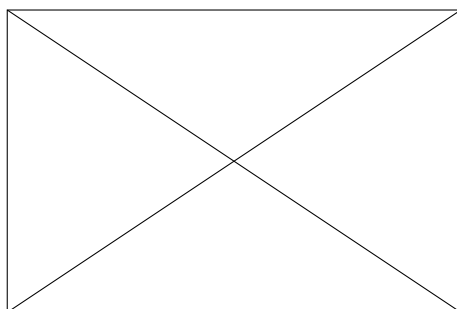
3.1 TikZ Paths

Inside a tikzpicture environment everything is drawn by starting a path and by extending the path. 

3.2 Example on Paths

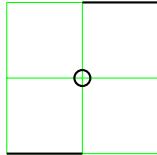


3.3 Defining Coordinate Labels

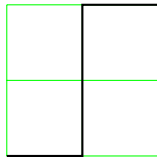


3.4 Path Extension Operations

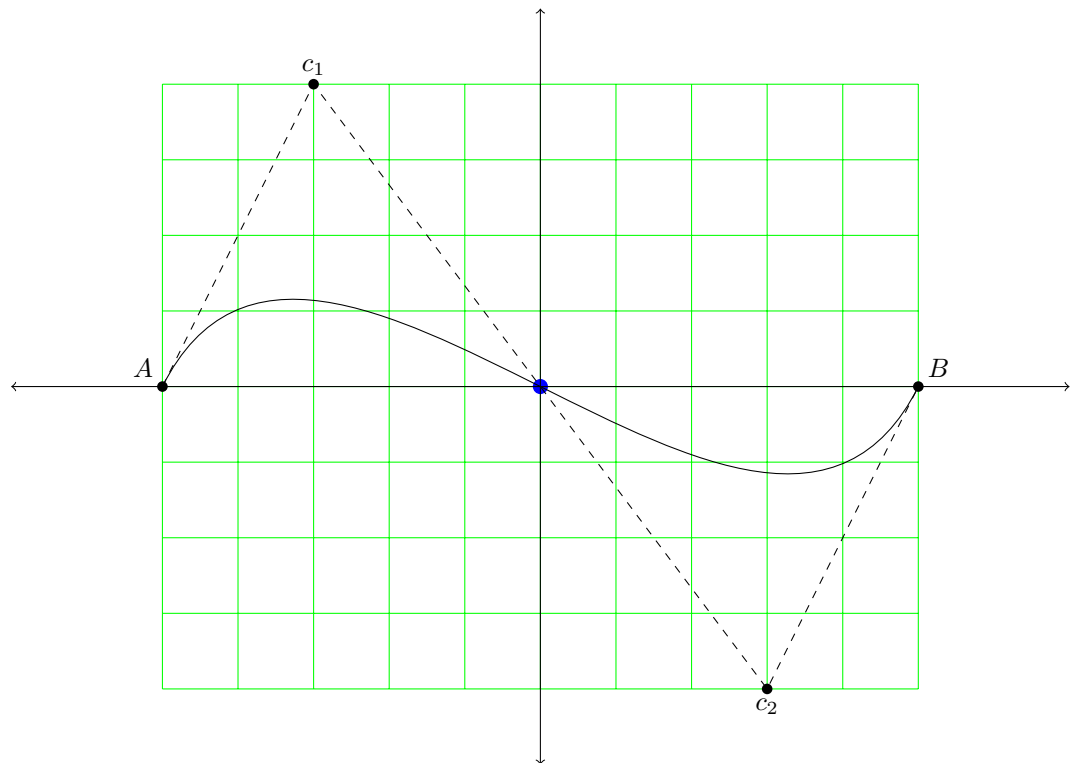
3.4.1 The move-to Operation



3.4.2 The line-to Operation



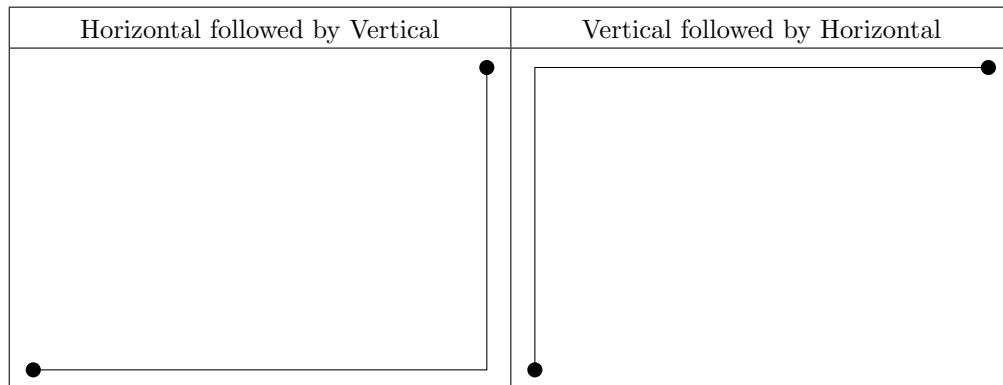
3.4.3 The curve-to Operation



3.4.4 The cycle Operation



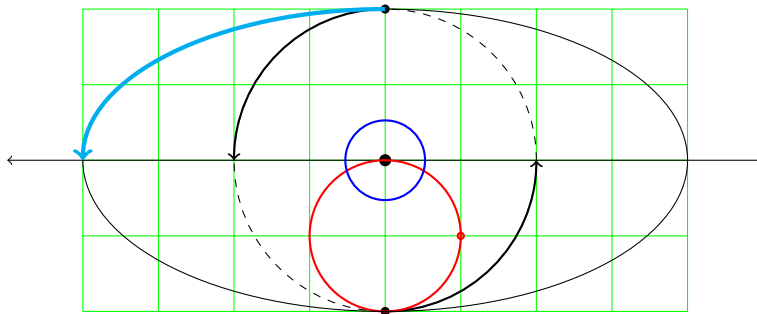
3.4.5 The Double line-to Operations



3.4.6 The rectangle Operation

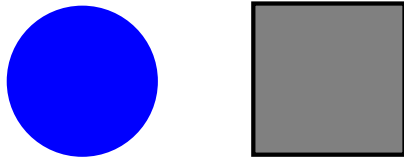


3.4.7 The arc and ellipse Operations



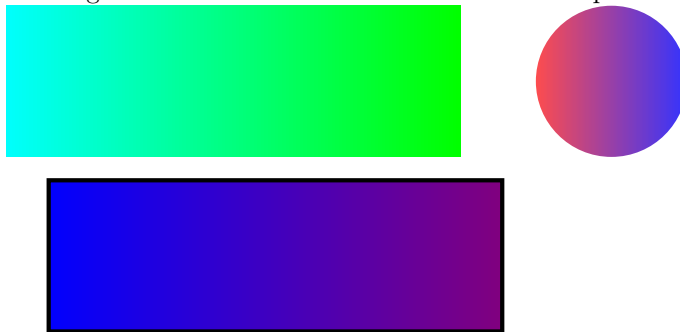
3.5 Styling Paths

3.5.1 Filling Paths

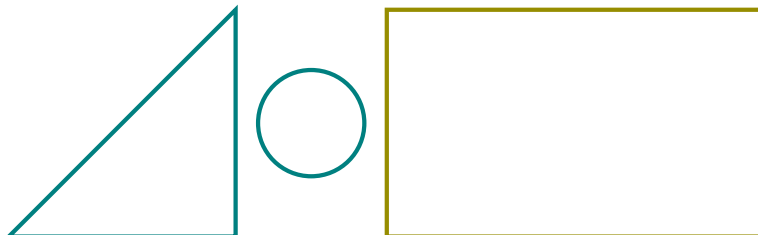


3.5.2 Shading Paths

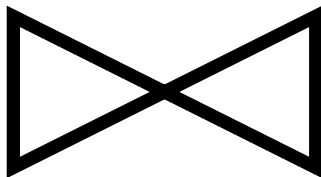
Shading Commands must be Inserted inside a `tikzpicture` environment.



3.5.3 Coloring Paths



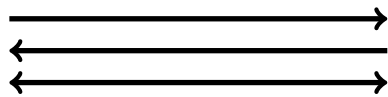
3.5.4 Setting the Line Width



3.5.5 Line Caps and Joins

Line Caps	Line Joins
round, rect, and butt	round, bevel, and miter
	

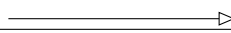
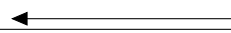
3.5.6 Arrows

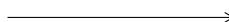


Changing the Arrow Head using >=

Style	Arrow Shape
Stealth	
Reversed	
to [def]	
Reversed	
latex	
Reversed	

Arrow Heads within *arrows* tikzlibrary

open triangle 90		triangle 90	
open triangle 60		triangle 60	
open triangle 45		triangle 45	

angle 90	
angle 60	
angle 45	

open diamond		diamond		o	
open square		square		*	

4 Nodes

So far we've been working with drawings that don't incorporate text or math or other form of content, and *the node operation* is the gateway to do so in tikz.

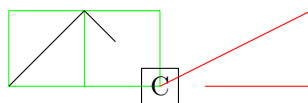
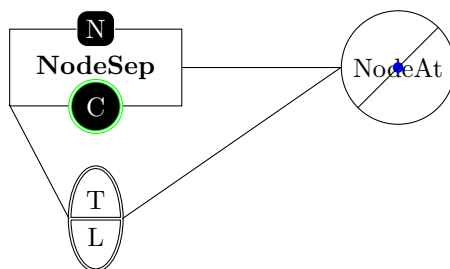
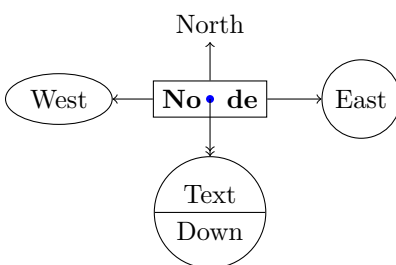
Base Node X_o^3

Table	Node
-------	------

Box Node

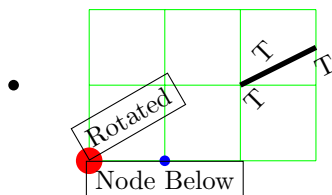
Circle

Small • Circle

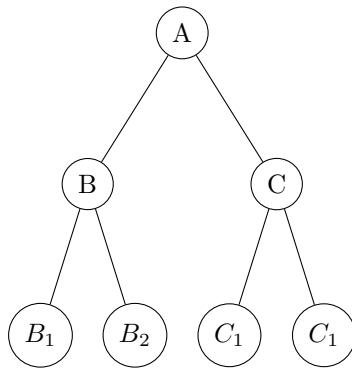
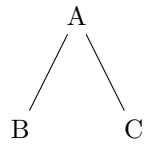


•

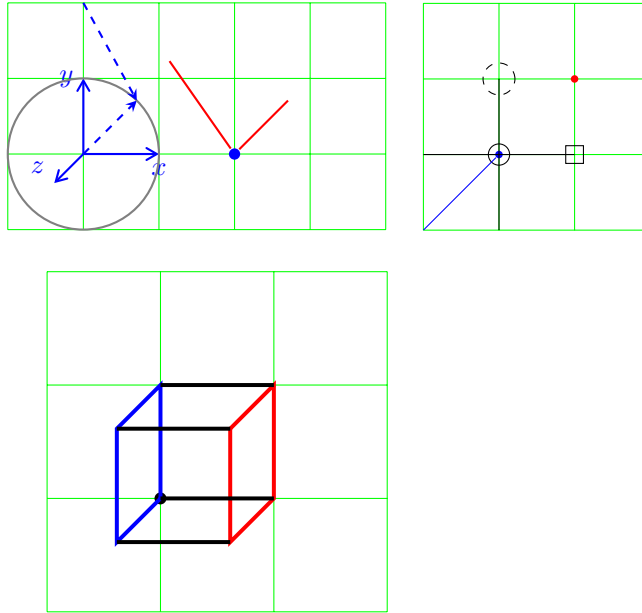
Anchoring



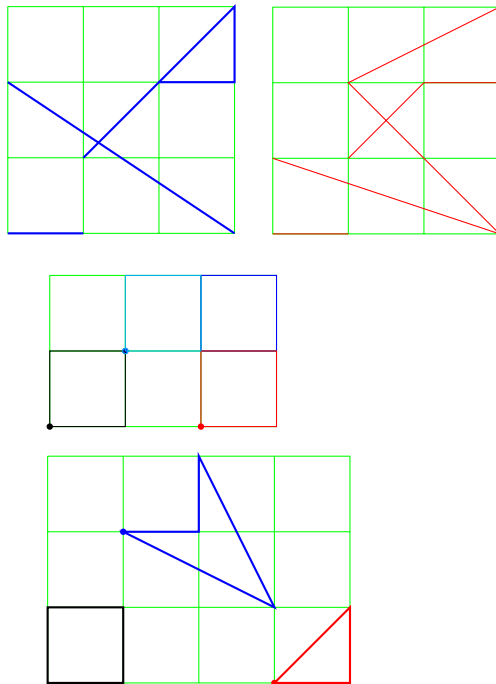
Constructing Trees



5 Understanding Coordinate Systems



5.1 Relative & Incremental Coordinates

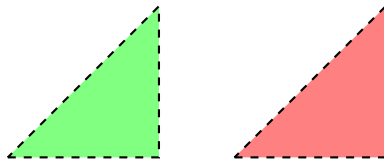


6 Defining Styles

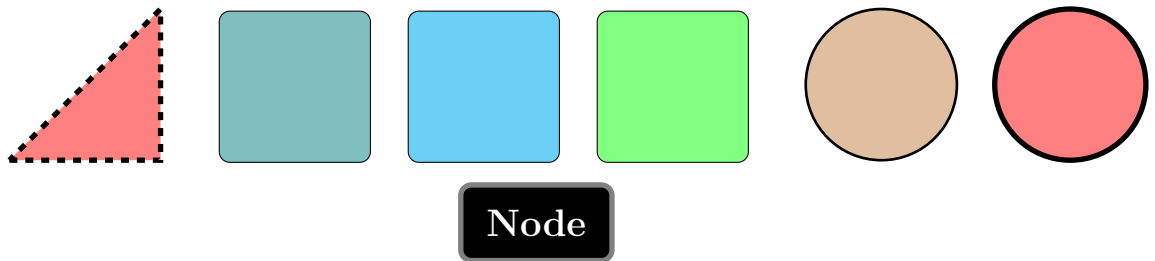
6.1 By Using the `tikzset` cmd – Globally



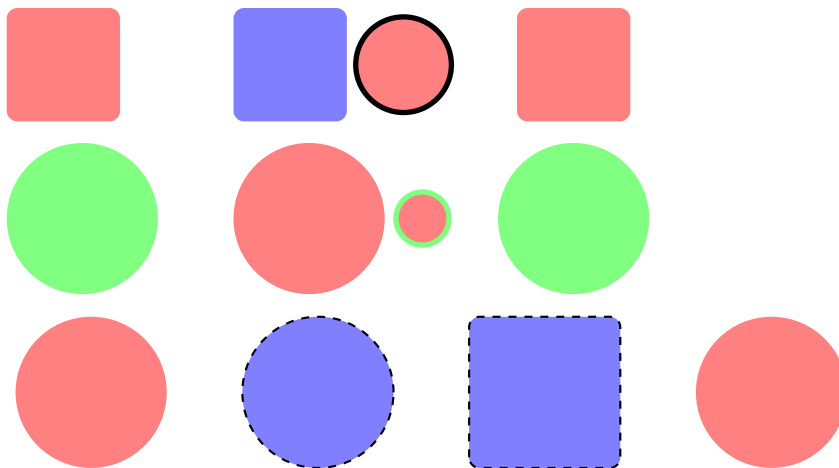
6.2 Inserting as a `tikzpicture` Option



6.3 Appending to an Existing Style

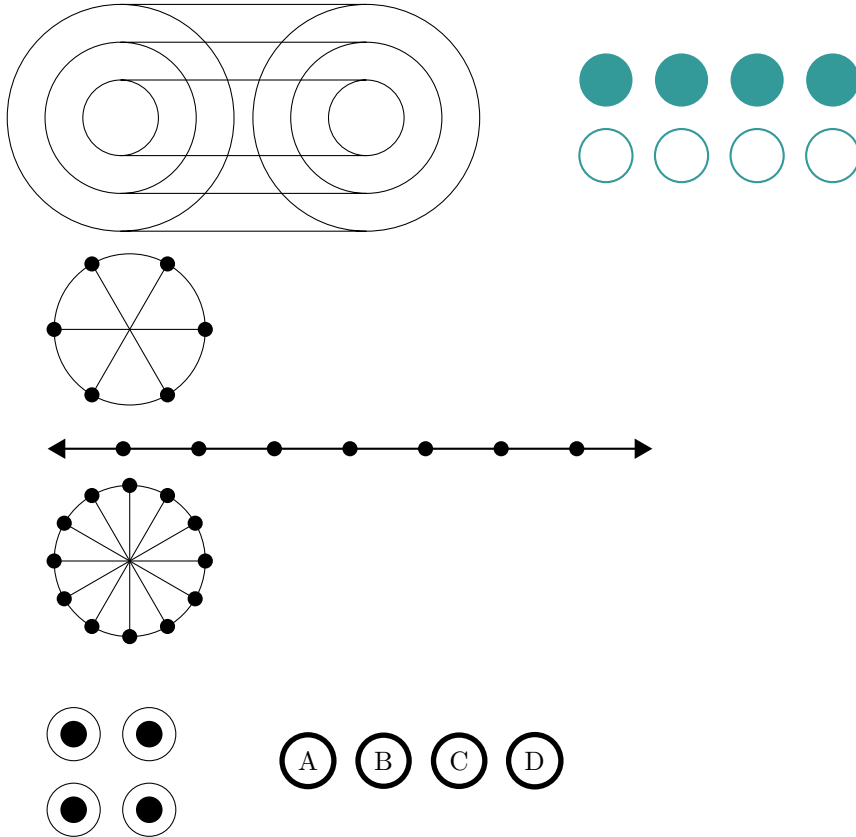


7 Scopes



8 Repeating Drawings

8.1 Using the foreach Command



8.2 Using the pic Command

